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IMPERIAL COLLEGE LONDON

DEPARTMENT OF COMPUTING

DLFuzz: Direct Input Generation with LLMs for Fuzzing

Project Report

Author:
Fares Shehata

Supervisor:
Prof. Cristian Cadar

Assistant Supervisor:
Bachir Bendrissou

Second Marker:
Prof. Alastair Donaldson

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Abstract

Software underpins almost every part of modern life, and the cost of software defects is enormous, running into trillions of dollars each year. Thorough testing is therefore essential, but writing tests that exercise the many corner cases of a complex program by hand is difficult and time-consuming. Fuzzing has become one of the most effective automated ways to find bugs: it runs a program on large numbers of generated or mutated inputs and watches for failures, and has uncovered critical vulnerabilities in compilers, operating systems, and other widely used software. How well a fuzzing campaign performs, however, depends heavily on the quality and diversity of the seed inputs it starts from.

At the same time, large language models (LLMs) have rapidly grown in capability and prevalence, and have begun to be applied to fuzzing - for instance, to synthesise input generators guided by code coverage. This report introduces DLFuzz, a tool that instead uses an LLM to generate seed inputs *directly*, evolving a pool of inputs under coverage feedback rather than evolving a program that produces them. We compare DLFuzz against the original generator-synthesis approach on two systems under test of differing complexity. We find that, although direct generation produces smaller, lower-coverage seed corpora, it reaches broadly comparable coverage once those seeds are handed to a downstream mutation fuzzer. We discuss why this happens, the limitations of the approach, and directions for making direct input generation more competitive.

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Chapter 1

Introduction

Testing software is important. Across the U.S. economy, the cost of poor software quality was estimated at at least \$2.41 trillion in 2022, with technical debt around \$1.52 trillion - illustrating the economic risk of defects and the value of rigorous verification and validation [1, 2]. However, maintaining a high-quality test suite is challenging: it is difficult to enumerate all relevant inputs, states, and corner cases, and infeasible to exercise every potential path through complex software systems. Systematic techniques like fuzzing help by exploring vast input spaces - including malformed or unexpected inputs that traditional test design may miss - and can be run continuously with runtime feedback (e.g., sanitizers, coverage) to catch regressions early [3, 4].

Fuzzing has proven to be a very successful technique to find bugs and vulnerabilities in code. Some of the most important software we use - language compilers, operating system components, cryptography libraries, and database systems - have had critical bugs discovered and fixed via fuzzing [3]. For example, Google’s large-scale fuzzing infrastructure has reported thousands of security vulnerabilities and tens of thousands of functional bugs across hundreds of open-source projects, leveraging coverage-guided engines (e.g., libFuzzer, AFL++) with sanitizers and scalable orchestration [5]. These systems rely on code-coverage guidance to steer execution toward new edges and behaviours; while coverage cannot literally guarantee “all code paths” are explored, it provides a practical, measurable heuristic to maximize path diversity during a campaign [5, 6].

There are two major categories of fuzzers: generation-based and mutation-based. Generation-based fuzzers use input formats or grammars to synthesize inputs from scratch that satisfy syntactic rules of the SUT. Mutation-based fuzzers begin with a corpus of example inputs (seeds) and iteratively mutate them [3]. In practice, grey-box mutation-based fuzzers often dominate modern vulnerability discovery because they (a) start from real inputs that already respect many syntactic/semantic constraints, enabling rapid progress past shallow checks, and (b) use lightweight program feedback (e.g., edge coverage, value profiling, branch distances) to adapt mutations toward novel paths and deeper states - especially effective for uncovering unexpected or undefined behaviour in complex code bases [3, 7].

Despite the strengths of mutation-based fuzzers, their effectiveness often hinges on the quality and variety of seed inputs. A narrow or unrepresentative corpus can trap a campaign in local minima, limiting path diversity and bug discovery; conversely, a rich, diverse seed set boosts pass rates through parsers and unlocks deeper program states [4]. Manually crafting such corpora is labour-intensive and time-consuming, and using existing samples “from the wild” offers no guarantee of exercising the full behavioural envelope of the SUT. As a result, many fuzzers invest in seed curation, deduplication, minimization, and dynamic corpus management to keep exploration efficient and targeted [7].

With the advent and widespread adoption of large language models (LLMs), their reasoning and contextual knowledge can alleviate several setbacks of existing fuzzing approaches. In mainstream software assurance, early evidence from large-scale industrial deployments shows that AI assistance can expand test generation and accelerate the on-boarding of new targets to continuous fuzzing; for example, Google reports that AI-generated fuzz targets and helpers have led to more vulnerabilities

being found, and expanded coverage across diverse ecosystems [6, 8]. In fuzzing research, LLMs have been used to generate grammars and structured inputs, to mutate inputs with semantic awareness, and to broaden language and domain coverage - spanning compilers, APIs, protocol/file formats, and more [9, 10, 11, 7].

Work has been done in using LLMs to generate fuzzers themselves, guided by coverage data [11], as well as generating inputs directly with a simple crash oracle [9, 12]. However, we have not seen any projects using LLMs to generate inputs, but also in a coverage guided manner. Our hypothesis is that using LLMs to generate and evolve seed inputs directly in a coverage guided manner will result in a more semantically and structurally diverse corpus, which will in turn lead to better fuzzing performance.

We present DLFuzz, a novel fuzzing approach that leverages LLMs to generate and evolve seed inputs directly in a coverage guided manner. DLFuzz uses an LLM to synthesize initial seed inputs that are semantically rich and structurally varied, and then iteratively mutates these seeds using the LLM, guided by coverage feedback from the SUT. This allows DLFuzz to explore deeper program states and uncover more complex bugs than traditional mutation-based fuzzers that rely on simpler mutation strategies.

Chapter 2

Background

2.1 Fuzzing

Fuzzing is an automated software testing technique that exercises a system under test (SUT) with large numbers of inputs - often randomly generated or systematically mutated - and observes the SUT for failures or other “interesting” behaviours [13, 14]. Historically, seminal studies by Miller and colleagues demonstrated that random inputs could reliably uncover robustness issues in UNIX utilities and services, popularizing the approach and coining the term “fuzz” [13, 14]. At minimum, a fuzzing campaign requires (i) a mechanism to obtain test inputs (e.g., via generation or mutation), and (ii) an oracle to judge whether an execution is interesting (e.g., crashes, sanitizer-detected undefined behaviour, coverage increases, or differential inconsistencies) [15, 16, 17, 18, 19, 20]. The term “fuzzing” encompasses multiple sub-categories: generation-based vs. mutation-based input creation, and black-box vs. grey-box vs. white-box feedback. These categories trade off generality, required access to the SUT, and effectiveness [14, 15].

Oracles can be defined in several ways. Classical “crash oracles” detect fatal signals or abnormal exits. Sanitizer oracles detect memory and concurrency errors at runtime - for example, AddressSanitizer (ASan) for buffer overflows and use-after-free [16], MemorySanitizer (MSan) for uses of uninitialized memory [17], UndefinedBehaviorSanitizer (UBSan) for undefined language behaviors [18], and ThreadSanitizer (TSan) for data races [19]. Coverage change can also serve as a meta-oracle to prioritize inputs that explore previously unseen code [15, 21]. Finally, differential testing oracles flag behavioural inconsistencies across multiple implementations or compiler/toolchain variants (e.g., Csmith-based cross-compiler testing, or Equivalence Modulo Inputs) [22, 20].

2.1.1 Generation-based fuzzing

Generation-based fuzzers synthesize inputs from scratch, often guided by formal grammars or input specifications, to produce syntactically valid (and sometimes semantically constrained) test cases [23, 24]. This approach is powerful when a grammar or model is available and correctness of the input structure is important. Representative systems include:

- **Csmith**, which generates well-defined C programs to differentially test compilers while avoiding undefined behaviour [22];
- **LangFuzz**, which composes code fragments guided by a grammar to stress language processors [24];
- **NAUTILUS**, which couples grammars with coverage guidance to explore deep states in structured input spaces [23].

A known challenge for generation-based fuzzing is probabilistic bias: the distribution over grammar productions (or construction rules) influences which features and paths are explored and which bugs are likely to surface. Certain early failures (e.g., parser rejects) may mask deeper behaviours. Design choices such as avoiding undefined constructs (as in Csmith) and integrating feedback into generation (as in NAUTILUS) mitigate these issues [22, 23]. Another limitation is domain

specificity: high-quality generators require grammar/specification knowledge and are not trivially reusable across unrelated SUTs [24].

2.1.2 Mutation-based fuzzing

Mutation-based fuzzers derive new inputs by mutating existing seeds, leveraging that real inputs satisfy many syntactic/semantic constraints and thus can bypass shallow checks [15]. Mutations range from low-level bit/byte flips and arithmetic on integers, to structural operators such as block insertion/deletion/duplication, splicing segments from multiple seeds, and dictionary/token-aware rewrites [21]. For example, in the AFL/AFL++ family of grey-box mutation fuzzers [25], the fuzzing loop is feedback-directed and proceeds roughly as follows:

1. Initialize a queue (corpus) from seed inputs.
2. Select a seed according to a power schedule that favours seeds with higher novelty/coverage potential, allocating them more mutation “energy” [15].
3. Apply a sequence of mutation stages, including deterministic bit/byte flips and small arithmetic tweaks, followed by randomized “havoc” (stacked mutations) and optional splicing with other seeds.
4. Execute the mutated input on an instrumented target using a low-overhead execution model (e.g., forking or persistent/in-process execution) to maximize throughput, with instrumentation added at compile time or via binary-level dynamic translation when source is unavailable [15, 26, 27, 21].
5. Record the coverage bitmap (e.g., edge coverage with hitcounts) and, if the input exercises previously unseen edges or yields a sufficiently “interesting” coverage change, enqueue it as a new seed and update scheduling statistics.
6. Triage, deduplicate (e.g., by coverage/signature), and minimize crashing inputs for efficient regression testing.

AFL++ extends this loop with numerous engineering improvements (e.g., richer mutation operators and scheduling heuristics, dictionaries, context-sensitive coverage, and persistent execution modes) [25] while retaining the same core principle: lightweight feedback steers mutations toward inputs that explore novel paths and uncover deeper bugs [15].

While highly effective, mutation-based fuzzing is sensitive to seed quality and diversity: a narrow seed set can trap the search in local minima, whereas diverse seeds improve parser pass rates and enable deeper exploration. Consequently, corpus management (seed curation, minimization/deduplication, and power scheduling) is a first-class concern in production fuzzers [15, 21].

2.1.3 Black-box, grey-box, and white-box feedback

Feedback-directed fuzzing adapts input generation/mutation based on information obtained from the SUT [15]. The extent and type of information distinguishes three broad modes:

- **Black-box** fuzzing executes the unmodified SUT and observes only external outcomes (exit codes, crashes, timing). It suits closed-source targets and can maximize throughput but offers limited guidance. Differential testing of compilers with Csmith is largely black-box with respect to compiler internals [22].
- **Grey-box** fuzzing uses lightweight instrumentation to obtain coarse-grained internal signals - most commonly edge or branch coverage, value profiling, and sanitizer findings - balancing guidance and throughput. Instrumentation can be added at compile time (e.g., compiler-based edge counters and sanitizer hooks, as in libFuzzer/Clang) or at the binary level via dynamic instrumentation (e.g., Pin) or dynamic translation/emulation (e.g., QEMU), with some performance cost [21, 27, 26].
- **White-box** techniques analyse code paths symbolically and use constraint solvers to generate inputs that satisfy path predicates; these are often considered symbolic or concolic execution rather than fuzzing per se. Classical systems include DART (directed automated random

testing), KLEE (systematic test generation for systems software), and SAGE (white-box fuzzing at scale) [28, 29, 30]. White-box approaches can be highly precise but often incur significant computational cost to manage path explosion and constraint solving.

2.1.4 Instrumentation, coverage, and sanitizers as oracles

Coverage instrumentation records which code elements are executed - line/statement coverage, branch coverage, edge coverage, and function coverage - providing both progress metrics and guidance for feedback-directed search [31]. In grey-box fuzzing, edge coverage (control-flow-graph edge hit maps) has become a de facto signal because it captures path diversity at low overhead. Compiler toolchains expose coverage counters (e.g., via sanitizer/coverage passes), and binary-level tools can approximate similar metrics [15, 27, 26]. Inputs that exercise previously unseen edges, or significantly alter the coverage profile, are prioritised for further mutation [15, 21].

Sanitizers enrich oracles by detecting classes of undefined or erroneous behaviour at runtime, also with compile time instrumentation.

- **AddressSanitizer (ASan)** detects invalid memory accesses (buffer overflows, use-after-free) with relatively low overhead. [16].
- **MemorySanitizer (MSan)** detects uses of uninitialized memory [17].
- **UndefinedBehaviorSanitizer (UBSan)** detects undefined behaviours such as signed integer overflows and invalid shifts [18].
- **ThreadSanitizer (TSan)** detects data races in multithreaded programs [19].

Collectively, coverage and sanitizers serve both as guidance signals and as oracles of “interestingness,” enabling fuzzers to triage and minimize crashing or sanitizer-violating inputs [15, 21].

2.2 LLMs

Large language models (LLMs) are neural networks trained to predict sequences of tokens and thereby generate, understand, and reason about text and code. The decisive architectural breakthrough was the Transformer, which replaced recurrence with self-attention, enabling efficient parallel training and better modelling of long-range dependencies [32]. From this foundation, families of models diversified: encoder-only models for representation learning (e.g., BERT) [33]; decoder-only, autoregressive models that excel at in-context learning and few-shot generalisation (e.g., GPT-3) [34]; and, at larger scales, systems such as PaLM that demonstrated gains from the triumvirate of data, parameters, and compute [35]. Alignment advances, notably instruction tuning and reinforcement learning from human feedback (RLHF), further improved models’ ability to follow natural instructions safely and reliably [36]. Open foundation models (e.g., LLaMA) broadened access and accelerated downstream innovation [37], while code-specialised LLMs (e.g., StarCoder and Code Llama) brought stronger capabilities to software engineering tasks [38, 39].

In software testing, LLMs have been adopted to reduce the cost of writing target-specific generators and harnesses, and to expand test diversity across languages and evolving features. For compilers and language processors, LLMs can automatically produce diverse, valid programmes that increase coverage and expose nuanced feature interactions [9, 40]. White-box flows integrate LLMs with compiler analyses to derive targeted test cases for particular optimisation passes [10]. Beyond test synthesis, LLMs help generate fuzz drivers and adapt prompts from documentation, cutting set-up time for continuous fuzzing [41, 42]. In lower-level domains, code-aware LLMs have been used to exercise deep kernel execution paths and uncover subtle defects [43]. Empirically, universal LLM fuzzers demonstrate significant coverage gains and numerous bug findings across compilers, SMT solvers, and quantum stacks, illustrating the breadth and productivity benefits of LLM-assisted testing [9, 12].

Nonetheless, naively sampling large models can yield low validity - especially in strict languages - and waste SUT cycles. Practical systems therefore combine LLM generation with principled loops that distil prompts, adapt generation strategies, and repair errors on the fly to raise the proportion of valid tests and sustain exploration [9, 12]. Looking ahead, hybrid designs that pair selective

feedback signals (e.g., simple coverage counters or domain oracles) with LLM breadth are likely to deepen path exploration further, while maintaining portability across languages and SUTs.

2.3 Literature Review

2.3.1 ELFuzz

ELFuzz proposes a new method of LLM-based fuzzing: instead of directly generating test inputs, it uses an LLM to synthesize generation-based fuzzers themselves, and then evolves those fuzzers using a coverage-guided notion of “fuzzer space”, until they become strong seed generators for downstream mutation fuzzers (e.g., AFL++). This decomposition turns the hard problem of writing a good generator into many small, LLM-friendly code edits with quantitative feedback at each step [11].

The motivation behind this approach is that building a high-quality generation-based fuzzer is difficult, as human-written grammars and semantic constraints are hard to author and maintain at scale. By generating and evolving a collection of fuzzers with an LLM, and ranking them by the coverage they achieve, the burden of manual specification is shifted to the model while progress is governed by an objective measure.

How ELFuzz works

ELFuzz implements an iterative three-stage evolution loop; each iteration “climbs” toward stronger fuzzers (Figure 2 and Figure 3 in the paper):

(1) LLM-driven mutation

Given a current pool of candidate fuzzers (small programs that emit inputs), ELFuzz asks the LLM to produce code-level mutants via three human-like edit operations:

- Splicing: combine head of one fuzzer with tail of another using fill-in-the-middle to glue the junction. This merges strengths of two candidates in one mutant.
- Completion: truncate a candidate at a random point and let the LLM complete the remainder, often adding missing logic and utility calls.
- Infilling: remove internal lines (“holes”) and let the LLM fill them while maintaining consistency with surrounding code.

These operations leverage modern LLM editing capabilities to produce syntactically valid, semantically meaningful variants that can plausibly generate more diverse and valid inputs than naive random code edits..

Ablation results show that removing the fuzzer-space model hurts the most, and among mutators, removing splicing causes the largest degradation (up to 26.2% on several benchmarks), consistent with its role in fusing complementary strengths.

(2) Fuzzer space exploration (coverage-based strength analysis)

Each mutant (and existing seed fuzzer) is executed briefly to approximate the “cover set” of code elements (edges) it can reach in the SUT.

Let $C(F)$ and $S(F)$ denote the cover set and strength of a fuzzer F respectively. ELFuzz defines a strength pre-order:

$$S(F_i) \sqsubset S(F_j) \iff C(F_i) \subset C(F_j)$$

Mutants strictly weaker than their seed parents (e.g., due to type errors or reduced coverage) are discarded; those that introduce new coverage are retained. Because different fuzzers can cover disjoint regions, some are incomparable (neither subset nor superset). This induces a lattice structure - the “fuzzer space” - over candidate generators. Crucially, it enables fine-grained comparisons that a single scalar (e.g., total coverage count) would obscure.

(3) Max-cover selection (elite picking to prevent pool blow-up)

To keep the population small and diverse, ELFuzz selects a fixed number (N) of elites whose union of cover sets is maximized. Let V be a pool of fuzzers, and $E \subset V$ be the subset which maximises the union of cover sets. $\mathcal{P}_N(V)$ represents all N -element subsets of V . Thus, we have:

$$E = \operatorname{argmax}_{S \in \mathcal{P}_N(V)} \left| \bigcup \{ \mathcal{C}(F) \mid F \in S \} \right|$$

A greedy approximation (with re-seeding heuristics) is used to avoid an exponential search, while still favouring elites that collectively expand coverage.

Implementation notes:

The design is LLM-agnostic. The paper uses a local CodeLlama-13B model for cost reasons, with short prompts that state the SUT and format hints in comments. The entire evolution controller and fuzzer-space analysis are lightweight Python (3.6 KLoC) compared to classic grammar pipelines (e.g., 80 KLoC for Csmith) [22, 11].

Coverage is approximated by executing a limited number of generated tests per candidate and recording edge coverage; this “quick probe” is sufficient to rank candidates and guide search while keeping the loop fast [11, 31].

Benchmarks and reported results

Subjects under test (seven real-world SUTs, up to 1.79 MLoC) include structured formats and interpreters: jsoncpp (JSON), libxml2 (XML), re2 (RegEx), CPython (Python), cvc5 (SMT-LIB2), SQLite (SQL), and libsvg (SVG). Baselines include grammar-based and constraint-based input generators (e.g., GLADE/ANTLR pipelines and ISLA constraints) translated or adapted as needed for fair comparison.

Input-generation effectiveness (RQ1):

ELFuzz’s synthesised fuzzers generate seed corpora with much higher coverage within 10 minutes; the paper reports up to 434.8% more covered edges than the second-best technique on cvc5, and consistent advantages across the other SUTs (Figures 7 and 8).

Bug-finding with AFL++ (RQ2):

Using ELFuzz seeds, AFL++ triggers more bugs and reaches milestones faster. On some targets (e.g., SQLite), ELFuzz seeds lead to up to 216.7% more injected bugs found within 24 hours, and lower time-to-25/50/75% of total bugs compared to the next best baseline (Table 6; Figure 9).

Real-world campaign (RQ2, cvc5):

Over 14 days with 30 AFL++ instances, the ELFuzz-seeded campaign found five previously unknown bugs; three were exploitable. Some were fixed upstream during the study (Appendix C).

Ablation study (RQ3):

Removing the fuzzer-space model (ELFuzz-NoFS) drops performance the most (gaps from 6.6% to 62.5% vs. the full system). Among mutators, removing splicing (ELFuzz-NoSP) yields the largest additional drop (up to 26.2%), while removing completion or infilling usually has smaller effects—consistent with splicing’s role in combining complementary behaviours.

Interpretable/extensible generators (RQ4):

Case studies on SQLite show synthesised fuzzers encode non-trivial features (e.g., command-line pragma handling, query sorting) and can be extended by external techniques such as Zest-style coverage-guided parameter evolution (Appendix E).

Cost and practicality:

Synthesis time is a one-off pre-fuzzing cost (hours, varying by SUT), after which the synthesised generators are fast to run. The paper discusses a trade-off: heavy synthesis can reduce “wall-clock” available for subsequent mutation fuzzing, but the improved seeds often outweigh this cost in coverage and bug-finding speed.

Limitations mentioned in the paper

- The prototype initializes from very simple random-byte seed generators for generality. Using stronger initial templates (or LLM-discovered protocol/grammar hints) could further improve convergence.
- If an SUT’s input language is poorly represented in the LLM’s training data (e.g., proprietary/binary formats), the LLM’s edits may lack the priors needed to synthesize good fuzzers. Prompt engineering or fine-tuning may be required, and purely binary formats remain challenging.
- ELFuzz currently optimizes the input-generation stage that seeds AFL++. Maintaining diversity continually throughout the entire mutation campaign (dynamic co-evolution of seeds) is not explored and is a direction for future work.
- On some targets, ELFuzz spends tens of minutes synthesizing seeds before the AFL++ phase; while seeds speed up bug discovery thereafter, the up-front cost can matter under tight time budgets.
- Results were achieved with a small local 13B parameter model. Larger models or tool-augmented prompting could improve edits, but at higher compute cost.

Takeaways and implications for our project

- The concept of "fuzzer space" provides an objective and quantifiable way to improve a set of generators. While this project will be working on seeds directly, using fuzzer space can also allow us to objectively quantify the strength of an input / set of inputs.
- Human-like code edits (splicing, completion, infilling) utilise the capabilities of the LLM to make intelligent and context-aware changes to generators. Again, we can apply this directly to seeds.
- Possible extensions mentioned in the paper that we could implement include: integrating semantic constraints when available (e.g., via constraint solvers or specs), running continuous co-evolution during the mutation phase, and exploring retrieval or fine-tuning for low-resource or proprietary domains.

2.3.2 Fuzz4All

Fuzz4All is a universal language fuzzing tool that aims to target a long-standing pain point in fuzzing: writing SUT-specific generators and mutation logic are brittle, labour-intensive, and hard to reuse across languages or fast-evolving features. Its core insight is to leverage two LLMs and an "autoprompting" stage to produce concise, task-specific prompts, then run an iterative fuzzing loop that uses examples and natural-language “generation strategies” to continuously create valid, diverse test programs and inputs for many SUTs and languages, without instrumenting the SUT [9].

How Fuzz4All works

Fuzz4All makes use of two LLMs, each for a different role:

- The **Distillation LLM** reduces verbose user inputs (language documentation or specification/examples) into an “input prompt” that succinctly captures what to fuzz and how (e.g., feature summaries, allowed forms, example shapes). In the paper, the authors use GPT-4 as the distillation LLM, as it is a large model tuned for complex instructional prompts. This is called the "autoprompting" phase

- The **Generation LLM** creates fuzzing inputs (programs, expressions, API calls, test cases) from the distilled prompt, combining natural language instructions with code snippets. Here they use the small local LLM StarCoder for generation, to reduce the inference cost associated with querying larger LLMs.

Autoprompting (distillation stage)

Given user inputs (target docs, feature descriptions, examples), the autoprompting algorithm produces and scores candidate prompts, returning the best one to start fuzzing.

Candidate generation:

Generating candidate prompts is split into two parts:

- First, a “greedy” prompt is used, querying the LLM with a temperature value of 0. This is tuned for obtaining a plausible prompt with high confidence.
- Then, sample several variants at higher temperature are queried to diversify phrasing / structure while staying faithful to the task.

Scoring:

Each candidate prompt is used to drive a short generation run, and a scoring function evaluates the prompt by the validity of its produced code with respect to the SUT’s acceptance predicate (e.g., “does the compiler accept this program?”). This is chosen as opposed to other possible metrics (e.g., coverage, bugs found) as valid inputs are most likely to exercise logic deep within the SUT, instead of just the frontend / input parsing phase.

Let p be a prompt, $\mathcal{M}_G$ be the generation LLM, c be a code snippet, and isValid be a function that returns 1 if the code snippet is valid with respect to the SUT. The scoring function is defined as:

$$\text{score}(p) = \sum_{c \in \mathcal{M}_G(p)} \text{isValid}(c, \text{SUT})$$

The prompt with the highest score is chosen to pass on the fuzzing loop stage.

Fuzzing loop (generation stage)

Fuzz4All’s loop iteratively updates the prompt and generates new inputs using three generation strategies, while a user-defined oracle flags bugs/crashes. A rolling "example" is stored in between iterations, and is updated using the generation LLM with one of the following strategies:

- **Generate-new:** instructs the model to create a fresh input from the distilled prompt alone.
- **Mutate-existing:** asks the model to modify the current example (mimicking mutation fuzzers), e.g., swap types, add computations, reorder constructs.
- **Semantic-equiv:** prompts the model to produce an input semantically equivalent to the example but syntactically different (useful for compilers and SMT where form changes trigger new internal paths).

At each iteration, Fuzz4All selects a valid input from the batch as the new example and randomly chooses one of the three strategies, concatenates the distilled prompt + example + strategy text, and queries the generation LLM to produce a new batch of inputs.

Each batch is passed to the SUT and a user-defined oracle detects “interesting” behaviours (e.g., crashes, assertion failures, differential mismatches). The loop continues until the time budget expires, returning the set of detected bugs.

Generation modes: general vs targeted fuzzing

Fuzz4All supports two different modes of execution, based on the original user inputs provided:

- **General fuzzing:** The distilled prompt summarizes the overall input language (e.g., GCC C docs, CVC5 SMT-LIB overview). The loop explores broadly, maintaining diversity by alternating strategies and continuously refreshing the example.

- Targeted fuzzing: The distilled prompt focuses on a specific feature (e.g., C keywords `typedef/union`, C++ `std::expected`, Go atomic/built-in libraries, Java keywords such as `instanceof/synchronized/finally`). This improves the “hit rate” on the target feature while still discovering off-target behaviors.

Benchmarks and reported results

Across six input languages (C, C++, SMT, Go, Java, Python) and nine SUTs (e.g., GCC/Clang, CVC5/Z3, Go toolchain, OpenJDK javac, Qiskit), Fuzz4All shows consistent advantages over SUT-specific baselines.

Coverage over 24 hours:

Fuzz4All achieves the highest end-of-run coverage on all studied targets, with an average improvement of 36.8% versus the best baseline. Per-target improvements include, +18.8% on g++ (vs. YARPGen), +24.9% on CVC5 (vs. TypeFuzz), +13.7% on Go (vs. go-fuzz), and +60.9% on Java (vs. Hephaestus). Fuzz4All avoids early plateaus by iteratively updating the prompt with examples and alternating strategies, sustaining discovery late into campaigns.

Validity and volume:

Fuzz4All often produces more diverse inputs but with lower average validity than strongly SUT-specific generators. Even with reduced validity, the net number of valid inputs and coverage gains are higher due to LLM breadth and loop diversity; GPU inference is the dominant bottleneck.

Targeted fuzzing effectiveness:

Fuzz4All achieves an average feature hit rate of around 83.0% across targeted campaigns. Targeted prompting yields significant increases in reaching specific features compared to general campaigns, with manageable cross-feature hits (sometimes beneficial for discovery).

Bug finding ability:

98 bugs were reported across studied SUTs, with 64 confirmed by developers as previously unknown. Examples include internal compiler errors in GCC triggered by nuanced feature combinations (e.g., `noexcept + std::optional`), segmentation faults in Clang, and crashes in Qiskit due to invalid circuit constructs unlikely to be explored by template-based generators.

Limitations mentioned in the paper

- More general prompts for strict languages (e.g., Go with static checks; SMT with tight theory constraints) can yield lower validity, requiring heavier filtering and oracle work. Fuzz4All still improves coverage, but wastes more SUT cycles on invalid inputs.
- The generation LLM (StarCoder in the study) is inference-bound, and fuzzing campaigns are bottlenecked by GPU resources, limiting throughput. Also, the autoprompting overhead (on average 3 minutes per campaign) is modest, but adds baseline time.
- Autoprompting relies on model priors over recent docs / examples. If documentation is very new, obscure, or under-represented (e.g., emergent quantum APIs), distillation and generation can miss edge cases or “hallucinate” constraints, reducing validity and targeted hit rates.
- Fuzz4All intentionally avoids instrumenting SUTs. While this helps practicality and portability, it limits access to fine-grained internal signals (e.g., coverage edges, value profiling) that grey-box mutation fuzzers exploit to accelerate deep exploration. The method compensates by prompt updates and strategy alternation, but some hard-to-reach behaviours may still benefit from hybridization with feedback-guided loops.

Takeaways and implications for our project

- Fuzz4All’s two-LLM pipeline (distillation + generation) enables practical, universal fuzzing across languages and SUTs without bespoke grammars or instrumentation, and makes best use of each LLM’s strength / cost trade-off effectively. A multi-LLM approach could be applied in our seed generation project.
- The fuzzing loop is a prompt-conditioned search: distil, generate, validate, then update the prompt with examples while rotating strategies to sustain diversity. This strategy of prompt and input evolution has proven effective, and we can use this as well.
- Possible improvements that Fuzz4All were not able to implement include: pairing prompt evolution with coverage information to further deepen SUT exploration.

2.3.3 LMFuzz

LMFuzz is a universal, code-generating fuzzer that tackles a key weakness of prior LLM-based fuzzers: low validity of generated programs. It introduces two complementary ideas: a multi-armed bandit formulation with Thompson Sampling (TS) to adaptively choose generation operators, and an LLM-driven program-repair loop that fixes compilation/runtime errors on the fly. Together, these components aim to preserve diversity while materially increasing validity and coverage across multiple languages and SUTs [12].

How does LMFuzz work

Prompt selection and scoring:

LMFuzz begins with a small set of prompt candidates P_i - the textual instruction given to the generation LLM that conditions what kind of code to produce for the SUT. For each prompt P_i , the generation LLM produces a batch of 30 programmes (code samples) $x_{i,1}, \dots, x_{i,30}$. Each programme is then executed against the SUT.

Let $\text{program_return}(x_{i,j})$ denote the SUT’s exit status when executing programme $x_{i,j}$. The prompt’s score is the number of programmes in its batch that run successfully (exit code 0):

$$\text{Score}(P_i) = \sum_{j=1}^{30} (\text{program_return}(x_{i,j}) = 0)$$

LMFuzz selects the prompt with the highest $\text{Score}(P_i)$ to seed the fuzzing loop; ties are broken by the prompts’ identifier order (smallest identifier wins).

Fuzzing loop:

On each iteration of the fuzzing loop, LMFuzz executes the following steps:

- Generate code with the “generative LLM” (StarCoder-1B in the study) from the current prompt.
- Execute the code; if errors occur, invoke the program-repair loop.
- Select a mutation operator via Thompson Sampling (see below) and mutate the prompt using operator-specific instructions.
- Save generated code and update the prompt for the next iteration.

Generation operators (three “arms”):

- **Generate:** ask the LLM to produce a new program from the current prompt (diversity-first).
- **Mutate:** request a modified version of the previously generated program (fine-grained edits, e.g., inserting functions, tweaking parameters).
- **Semantics:** ask for a semantically equivalent program expressed differently (syntax changes that preserve behaviour), useful for compilers/solvers.

Operator selection via Thompson Sampling

LMFuzz treats operator choice (Generate/Mutate/Semantics) as a multi-armed bandit problem. For each operator, it tracks successes and failures (e.g., “compiled and ran without error” vs. not) and samples a score from a Beta posterior:

Arm k maintains counts α_k (successes) and β_k (failures) and draws

$$\theta_k \sim \text{Beta}(\alpha_k, \beta_k)$$

The operator with maximal sampled θ_k is selected for the next iteration, balancing exploration and exploitation under uncertainty [44]. This probabilistic selection improves over random choice by favouring operators that currently yield valid, high-coverage code while still probing alternatives.

Program-repair loop (LLM-based error correction)

Generated programs frequently include basic issues (missing headers/imports, syntax errors, incorrect API usage). LMFuzz addresses this with a second “correction LLM” (DeepSeek-Coder-6.7B in the study). If the program fails, LMFuzz creates a “code+error” pair by concatenating the original source with compiler/runtime error messages and feeds it to the correction model.

The correction output replaces the program when valid; otherwise, LMFuzz can perform an additional repair cycle (the study evaluates 0/1/2 cycles). One repair cycle typically yields the best trade-off between validity gains and overall coverage (too many repair cycles reduce throughput and program diversity).

Benchmarks and reported results

Coverage versus baselines:

Across GCC, G++, CVC5, Go, and Qiskit, LMFuzz achieved higher or competitive line coverage than both traditional fuzzers (e.g., GrayC, Csmith, TypeFuzz, go-fuzz, MorphQ) and Fuzz4All. Examples from Table 3:

- GCC: LMFuzz 199,595 lines, coverage rate 16.57% (higher than GrayC 13.90% and Csmith 9.27%).
- G++: LMFuzz 223,129 lines, 19.08% (higher than YARPPGen 14.25%).
- CVC5: 48,713 lines, 9.12% (higher than TypeFuzz 8.64%).
- Go: 38,024 lines, 12.03% (higher than go-fuzz 11.32%).
- Qiskit: 31,565 lines, 19.63% (slightly higher than Fuzz4All 19.45%, competitive with MorphQ 20.39%).

Validity of generated code

LMFuzz’s code validity is roughly twice that of Fuzz4All across five projects (e.g., C/C++ validity in the 60–70% range for LMFuzz vs. 30–37% for Fuzz4All), though still below specialized, traditional generators with strict grammars.

Bug finding ability:

24 bugs reported across five SUTs: GCC (4), Clang (9), CVC5 (2), Qiskit (9); 7 previously unknown confirmed by developers and 17 known/rediscovered (Table 7, Fig. 9).

Ablations:

- Thompson Sampling vs random selection: TS yields higher line coverage across all subjects (Table 4).
- Repair cycles: one cycle improves validity markedly; two cycles further increase validity but reduce coverage/diversity (Table 5).

- Prompt-quality scoring: higher-scored prompts lead to better validity and coverage, validating the initial scoring function.

Limitations mentioned in the paper

- While validity improves substantially over prior LLM fuzzers, it remains below strictly grammar-driven generators. More repair cycles can raise validity but reduce diversity/coverage due to time budget constraints.
- Performance depends on initial prompt quality and model priors; the scoring function helps, but prompts that omit critical headers/imports or feature details degrade validity (Chapter 4.2.3). Updates to foundation models pose external validity threats.
- The approach is inference-bound; per-iteration generation and repair consume GPU time, limiting the total number of programs explored under fixed budgets.
- Evaluation uses line coverage (via project tools or repository line counts). Without grey-box instrumentation (e.g., edge/branch/value profiling), some deep behaviours may remain under-explored compared to feedback-guided mutation fuzzers.
- The TS bandit selects among three operators. Extending to richer operator families (grammar-aware rewrites, API-sequence planners) may improve generality but complicates posterior credit assignment and sampling dynamics.

Takeaways and implications for our project:

- Modelling operator choice as a bandit with Thompson Sampling gives a principled and effective way to steer LLM generation toward valid, high-coverage programs under uncertainty [12, 44].
- A lightweight, in-loop program-repair mechanism materially boosts validity and practical bug-finding without hand-built grammars.
- For new LLM fuzzers, combining adaptive operator selection, prompt-quality scoring, and on-the-fly repair appears crucial to escape the low-validity trap while retaining diversity; hybridizing with selective feedback signals (e.g., simple coverage counters) could further deepen exploration.

Chapter 3

DLFuzz

We present DLFuzz, a modification of ELFuzz [11] that uses the LLM to directly synthesise test inputs, rather than to synthesise the generation-based fuzzers that produce them. ELFuzz solves the problem of writing a good input generator by delegating it to an LLM and steering the search with coverage feedback over a so-called *fuzzer space*. Our hypothesis is that the same machinery (LLM-driven mutation, coverage-based strength comparison, and elitist selection) can be applied one level lower, to the inputs themselves. The result is a coverage-guided, LLM-driven seed-corpus synthesiser: it produces a diverse, high-coverage set of seed inputs that can be used as a corpus in their own right or handed to a downstream coverage-guided, mutation-based fuzzer such as AFL++ or libFuzzer [25, 21].

This chapter describes the design and implementation of DLFuzz. The development of DLFuzz was in two major stages. The first stage was a direct replacement of a *generator* with a *collection* of inputs, to most closely preserve the structure of the original pipeline. After discovering issues with this design, the second stage removes collections entirely, and maintains a single evolving population of inputs, as this is a more natural implementation.

3.1 From Generators to Inputs

ELFuzz evolves a population of *generators*: small programs (in the prototype, Python functions) that, when executed, emit a test input for the SUT. Evolution proceeds as an iterative loop in which each generation is mutated by the LLM, the mutants are assessed by the coverage they reach, and the strongest are retained as seeds for the next generation [11]. A crucial property of this design is that the unit of evolution, a generator, is itself a *source* of inputs: a single generator can be executed arbitrarily many times to produce an unbounded number of distinct inputs. Coverage of a generator is therefore measured by running it a fixed number of times and taking the union of the edges its outputs reach, approximating its *cover set* $\mathcal{C}(F)$.

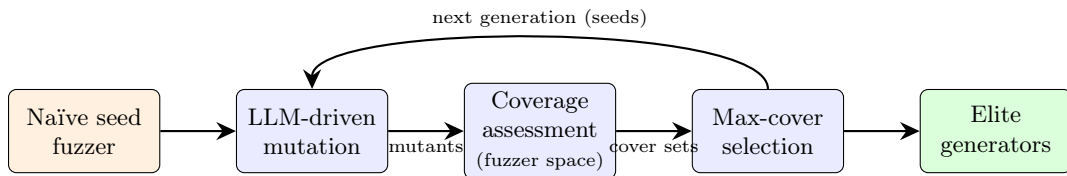


Figure 3.1: The ELFuzz evolution loop, which DLFuzz adapts. Each iteration mutates the current generators with an LLM, places the mutants in the fuzzer space according to their cover sets, and selects a fixed number of elites that maximise the unioned coverage.

Our project removes the generation step entirely. Instead of asking the LLM to write a program that emits an input, we ask it to emit the input. This eliminates a layer of indirection and the associated cost of executing generators to obtain their outputs. It also sidesteps the need for the LLM to produce syntactically valid *code* in a fixed host language, as it only needs to produce

content in the SUT’s input format. It also raises an immediate design question: in ELFuzz, the unit of evolution is a generator, but a generator is not the same kind of object as an input. A generator stands in for a whole family of inputs, whereas a single input is just one point in the input space. The two designs that follow differ precisely in how they answer the question of what the unit of evolution should become once generators are removed.

3.2 Stage One: Replacing Generators with Collections of Inputs

The most direct translation of ELFuzz would replace each generator with a single input and reuse the rest of the pipeline unchanged. However, this is not an equivalent design. A generator can produce an unbounded number of inputs, and therefore can reach many parts of the SUT. A single input exercises exactly one path through the SUT, so can only ever cover a small, fixed portion of it. Substituting one input for one generator would thus replace a rich object with a far weaker one, and the coverage-driven search would have far less to work with at each point in the population.

To preserve the expressive power of the unit of evolution, our first design replaces each generator with a *collection* of inputs. A collection plays the role that a generator played in ELFuzz: it is the object that is mutated, assessed, and selected. Its cover set is the union of the cover sets of its member inputs, so a collection of many diverse inputs can reach as much of the SUT as a capable generator, while remaining a concrete corpus of seeds rather than a program.

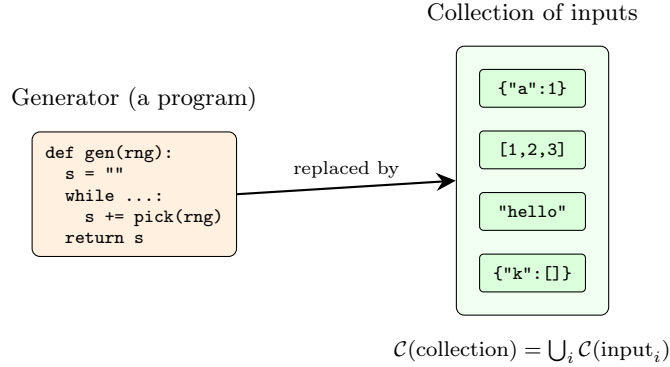


Figure 3.2: Replacing a single generator with a collection of inputs. The collection’s cover set is the union of its members’ cover sets, so it preserves the role the generator played as the unit of evolution.

3.2.1 Mutating individual inputs

We reuse ELFuzz’s three LLM-driven mutation operators - splicing, completion, and infilling - but apply them to individual inputs rather than to generator source code. In ELFuzz these operators act on programs and are realised through fill-in-the-middle (FIM) queries to the model [11]; here they act on the raw bytes of an input, cut at random line boundaries:

- **Completion:** an input is truncated at a random line boundary and the LLM is asked to continue it, extending the input with new, contextually consistent content.
- **Infilling:** two random line boundaries are chosen, removing a middle portion of the input, and the LLM fills the resulting hole via a FIM query while keeping the surrounding content coherent.
- **Splicing:** a prefix taken from one input and a suffix taken from another are joined, and the LLM produces glue content to fuse them into a single valid input. As in ELFuzz, splicing is the operator that combines the strengths of two parents.

These byte-level edits are the input-space equivalents of ELFuzz’s line-level edits over generator code. Where the original implementation cuts and rejoins lines of a Python program, we cut and

rejoin lines of, for example, a JSON document or an SQL script. The mutated input is always assembled from the preserved prefix and suffix plus the LLM-generated bridge, so that even when the model produces poor content the surrounding structure of the parent is retained.

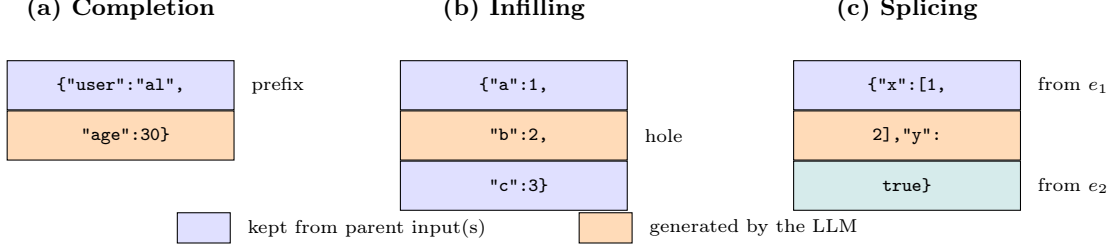


Figure 3.3: The three mutation operators applied to JSON inputs. Shaded regions are taken verbatim from the parent input(s); the remaining region is generated by the LLM to bridge the cut. Splicing joins a prefix of one parent (e_1) to a suffix of another (e_2).

3.2.2 Selecting elite collections

Because mutation continually produces new candidates, the population must be pruned to prevent it from growing without bound. ELFuzz achieves this with *max-cover selection*. From the pool it retains a fixed number N of elite generators whose unioned cover set is maximal, approximated greedily [11]. The collection-based design applies exactly the same principle, one level lower. Each collection is scored by its unioned cover set, and the strength pre-order

$$\mathcal{S}(C_i) \sqsubset \mathcal{S}(C_j) \iff \mathcal{C}(C_i) \subset \mathcal{C}(C_j)$$

ranks collections by the subset relation between their cover sets, just as ELFuzz ranks generators. We retain the N collections whose union of cover sets is largest and discard the rest, keeping the population at a fixed size while favouring elites that collectively expand coverage.

3.2.3 Limitations of the collection-based design

Although this design maps cleanly onto the ELFuzz pipeline, evolving collections rather than generators introduces two problems that do not arise when the unit of evolution is a true generator.

The corpus cannot grow. The population is a fixed number of collections, each holding a fixed number of inputs, so the total number of seeds is bounded throughout the run. With generators this bound is harmless, because each generator can produce an unbounded amount of inputs after evolution finishes. With collections the bound is real; the corpus can never contain more inputs than the initial configuration allows, regardless of how much new coverage the search discovers, and an input can only enter a collection by displacing another.

Elitism discards useful inputs. The collections are disjoint, but coverage is a property of individual inputs, and useful inputs are scattered across collections. Suppose one collection contains an input that uniquely covers some region of the SUT, and a different collection contains an input that uniquely covers another, disjoint region. Max-cover selection chooses whole collections, so if only one of the two collections is retained as an elite, the valuable input in the other is discarded along with it. The unit of selection (a collection) is coarser than the unit that actually carries coverage (an input), so selecting elites loses information and discards genuine progress.

3.3 Stage Two: Evolving a Single Population of Inputs

Both limitations above stem from the same root cause: coverage lives on individual inputs, but the collection-based design forces selection and accounting to operate on coarser groups. The second design removes the mismatch by making the input itself the unit of evolution. Rather than many disjoint, fixed-size collections, we maintain a single growing population - a *pool* - of inputs, and

every decision is made per input (Figure 3.4). This is both conceptually simpler and strictly more expressive than the collection-based design.

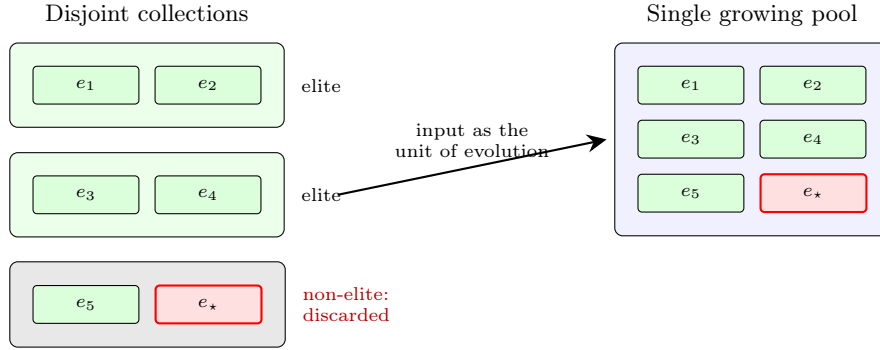


Figure 3.4: From many disjoint collections (left) to a single growing pool (right). On the left, the uniquely valuable input e_* is trapped in a non-elite collection and discarded with it; on the right it survives. Treating the input as the unit of evolution lets every input that contributes coverage survive, and lets the corpus grow without a fixed bound.

3.3.1 Initialising the pool

The evolution loop needs an initial pool to mutate. ELFuzz begins its own evolution from a single *generic generator*: a naive seed fuzzer that emits purely random bytes. It relies on LLM-driven evolution to gradually turn that noise into a structured generator. The paper notes that stronger initial templates could improve convergence [11]. Our first design followed the same spirit at the input level: the local LLM was queried to produce the entire starting population from scratch, each input generated independently from a short prompt naming only the target format.

In practice this proved a poor fit for the code-completion models used in the evolution loop. Such models are trained to *continue* existing text or code given substantial surrounding context. When prompted to produce a complete, well-formed input of a given format from a one-line description, they frequently returned empty, truncated, or malformed outputs that the SUT rejected outright and contributed no coverage. A from-scratch prompt gives the model too little to anchor on, and a small local model has weak priors for emitting a valid document of an arbitrary format unprompted.

An intermediate design seeded the pool from a *seed corpus* of real example inputs: small, publicly available samples such as JSON documents, XML files, or SQL scripts. This was so that the population began as a set of guaranteed in-format, coverage-bearing inputs. Where the local LLM was still used to generate further inputs it was primed with a few corpus examples (few-shot prompting) so it imitated real structure rather than inventing one from nothing. This improved the quality of the initial pool, but introduced a dependency on the availability of a suitable corpus for each target format, and the diversity of the starting population remained limited by whatever the corpus happened to contain. Also, the coverage achieved by the initial corpus was effectively free, meaning evaluating our approach against ELFuzz, which does not have this initial coverage advantage, would be unfair.

The final design replaces the real-world corpus with a small set of inputs *pre-generated* by a different, more capable model: Claude Opus 4.8 [45]. The model is given a single instruction naming the target format and asking for a fixed number of inputs that are each semantically and structurally distinct from one another. For example, the prompt used for cpython3:

Please generate five long semantically varied Python programs that utilize many features of the language. These inputs shall serve as a seed corpus for a fuzzing procedure that evolves these inputs. The programs do not need to perform any meaningful function, but they should exercise the scope of the language.

Before evolution begins, the powerful model was prompted to produce a fixed number of inputs that are each semantically and structurally distinct from one another. This closely mirrors how ELFuzz

initialises its own search: just as ELFuzz provides a single Generic generator as the shared base from which all generators evolve, our pre-generated inputs serve as the fixed starting population from which all subsequent mutations depart.

The split between two different models is deliberate. The local code-completion models used in the evolution loop excel at extending and recombining existing text, exactly what the mutation operators require. However, they are not well suited for generating varied, semantically interesting documents from scratch, a task that demands a strong prior over the target format and the capacity to produce meaningfully different examples on each call. A large frontier model such as Claude Opus 4.8, prompted to generate k inputs that are each semantically distinct from one another, reliably produces a diverse, well-formed starting population. Because this pre-generation step is performed once per target and its outputs are stored, the cost is paid only at setup time and does not recur across the evolution loop.

3.3.2 The evolution loop

The evolution loop mirrors ELFuzz’s three stages, but operates on inputs. The pool begins with the pre-generated inputs described in Chapter 3.3.1, after which each generation performs three steps:

1. **Mutation.** A batch of candidate inputs is produced by repeatedly drawing one (or, for splicing, two) inputs at random from the pool and applying one of the three mutation operators described in Chapter 3.2.1.
2. **Coverage assessment.** Each candidate is executed on the SUT and its cover set - the set of edges it reaches - is recorded.
3. **Curation.** Each candidate is compared against the current pool and either added, used to replace an inferior input, or discarded, according to the decision rule below.

3.3.3 Curating the pool

Curation is where the strength pre-order of the fuzzer space is enforced at the level of individual inputs. We say an input with cover set e_1 is *superior* to one with cover set e_2 when it strictly dominates it: $e_2 \subset e_1$. This is the input-level instance of ELFuzz’s relation $\mathcal{S}(F_i) \sqsubset \mathcal{S}(F_j) \iff \mathcal{C}(F_i) \subset \mathcal{C}(F_j)$. Letting U denote the union of the cover sets of all inputs currently in the pool, each candidate with cover set e is handled as follows:

- If the candidate covers edges not yet in the pool ($e \setminus U \neq \emptyset$), it is added to the pool. Any existing input that the new one strictly dominates is then evicted, since its coverage is now redundant.
- Otherwise the candidate reaches no new edges, but it may still be a better representative of coverage the pool already has. If it strictly dominates some existing input, it replaces the input it dominates by the largest margin.
- Otherwise the candidate is redundant with respect to the pool and is discarded.

A candidate that reaches no edges at all - for example, a malformed input the SUT rejects immediately - is always discarded. We additionally break ties by input size: when a candidate covers exactly the same edges as an existing input but is smaller, it replaces it. This is a deliberate departure from ELFuzz, whose selection rewards coverage with no size penalty; for a *seed corpus*, smaller inputs that achieve the same coverage are preferable, because they mutate and execute more cheaply in a downstream fuzzer.

The effect of this rule is that the pool always retains the union of all coverage discovered so far while continuously pruning inputs whose coverage is achieved by others. It approximates a Pareto frontier of inputs under the coverage-subset relation, refined by the size penalty, analogous to the max-cover elites of ELFuzz, but it is maintained incrementally and without a fixed cap on its size.

3.3.4 Properties

This design resolves both limitations of Chapter 3.2.3. Because selection now operates on the same unit that carries coverage, no progress is lost. Two inputs with disjoint coverage both survive, as each contributes edges the other lacks. Additionally, because the pool is a single set with no fixed capacity, the corpus is no longer capped by an initial configuration; it admits every input that contributes new coverage.

One consequence of our design is that the size of the pool does not necessarily increase monotonically. When a candidate introduces new edges, it is added. But, it may simultaneously dominate several existing inputs at once, all of which are then evicted. The net effect can be a smaller pool than before the candidate arrived. Crucially, however, eviction only ever removes an input whose coverage is wholly contained in that of a retained input, so the union of cover sets (the total coverage held by the pool) never decreases. Overall coverage is therefore monotonically non-decreasing across generations, even though the population size is not. As the search discovers individual inputs that each cover what previously took several, the corpus can become both smaller and stronger, with the size tie-break of Chapter 3.3.3 reinforcing the same trend. This is exactly the behaviour we want from a seed corpus, where fewer, higher-coverage seeds make for a more efficient downstream fuzzing campaign.

3.4 Implementation

DLFuzz is implemented as an additional *direct* mode within the existing ELFuzz codebase, reusing its configuration, coverage tooling, and LLM-serving infrastructure wherever possible. The pipeline is driven by two shell scripts that orchestrate a small number of Python components.

Orchestration. An outer script bootstraps the initial pool from the pre-generated inputs, placing them into the pool, computing their coverage, and retaining those that cover any edges. Then, it runs the per-generation loop for a configured number of generations, or until an optional wall-clock budget is exhausted. Each generation is handled by an inner script that performs the three steps of mutation, coverage assessment, and curation. To reuse the existing per-generator coverage tooling unchanged, each candidate input is written into its own directory so that the tooling treats it as an independent unit and reports its coverage separately.

Initialisation and mutation. Initialisation is handled by a component that reads the inputs pre-generated by Claude Opus 4.8 and places them into the pool, with each candidate assessed for coverage and discarded if it reaches no edges. Mutants are produced by a second component that draws inputs from the pool and applies the completion, infilling, and splicing operators; the choice of operator per candidate is random over the enabled set, and any operator can be disabled to support ablation studies.

Inputs tend to grow over successive generations: completion appends new content, and each generation’s outputs become the next generation’s parents, so left unchecked the parent text handed to the model would expand without bound. To keep every query bounded, the parent text fed to the model for an operator is truncated to a fixed size, snapped to the nearest line boundary. This truncation affects only the model’s view; the full, untruncated parent is still used when assembling the final candidate, so the input itself is never clipped.

Curation. The decision rule mentioned above is implemented by a curator that ingests the per-candidate coverage map and the current pool index, applies the add/replace/discard rule to each candidate, and writes the updated pool together with a record of the action taken for each candidate. The pool is stored as a flat directory of inputs alongside an index mapping each input to its cover set and size, so that the union U and the domination checks can be evaluated efficiently each generation.

Chapter 4

Evaluation

The aim of this project is to test the hypothesis laid out in the introduction (Chapter 1). ELFuzz produces seed inputs *indirectly*: it uses an LLM to synthesise generation-based fuzzers, and those fuzzers are then run to emit the inputs [11]. DLFuzz removes the intermediate program and asks the LLM for the inputs directly. Our hypothesis is that this is a more direct route to diverse, semantically interesting inputs, and therefore to more varied coverage of the system under test (SUT) - which is ultimately what a fuzzing campaign is trying to achieve. By eliminating the generator, the model’s knowledge of the input format is spent on the inputs themselves rather than on writing programs able to produce them, and coverage feedback selects directly over the objects we care about.

This chapter sets out how we tested that hypothesis. The whole evaluation is a direct comparison between original ELFuzz and DLFuzz, run under identical models, hardware, and time budgets:

4.1 Aims and Research Questions

We split evaluation into two research questions.

- **RQ1 (seed quality).** How much SUT coverage does each tool’s seed corpus achieve *on its own*, before any mutation fuzzing? This measures the raw quality of what each approach produces, and is the most direct test of the hypothesis: if directly generated inputs are more varied, they should cover more of the SUT.
- **RQ2 (downstream fuzzing).** When the seed corpus is used to bootstrap a grey-box mutation fuzzer (AFL++ [25]), how does coverage grow over the campaign, and where does it end? A more diverse corpus should let the fuzzer past shallow checks sooner and reach a higher plateau.

4.2 Experimental Setup

4.2.1 Systems under test

We evaluate on a subset of the SUTs used by ELFuzz, spanning structured data formats and language interpreters [11]. Table 4.1 lists the two SUTs evaluated so far with their input format and the size of their AFL++ coverage map. We deliberately chose targets at opposite ends of the complexity range: jsoncpp, a small JSON parser, and cpython3, the full CPython interpreter, which is roughly $25\times$ larger by instrumented edges.

4.2.2 Configurations compared

The two configurations are ELFuzz and DLFuzz. To isolate the effect of the design difference, every other factor is held fixed:

SUT	Input format	Instrumented edges
jsoncpp	JSON	5,236
cpython3	Python	133,395

Table 4.1: The systems under test evaluated so far (a subset of the seven ELFuzz SUTs). “Instrumented edges” is the size of the AFL++ coverage map (the total reachable edges of the target), used here as a direct measure of complexity.

- **LLM:** Qwen2.5-Coder-1.5B [46], a small local model chosen to fit the available GPU. Both tools use the same model for their evolution loop.
- **Hardware:** Intel i9-13900K CPU, 32GB RAM, NVIDIA RTX 4070 Ti GPU.
- **Fuzzer:** Both pipelines end by feeding their seed corpus to the same AFL++ configuration.

4.2.3 Testing Procedure

Each SUT is run end-to-end through the same four-stage pipeline for both tools:

1. **Synthesis / evolution (6 h).** ELFuzz evolves a population of generators; DLFuzz evolves the input pool of Chapter 3. Both are given a 6-hour wall-clock budget.
2. **Corpus production.** For ELFuzz, a separate **produce** step runs the evolved elite fuzzers for **200 seconds in total**, which is already sufficient to emit 1 million candidate inputs. DLFuzz has no **produce** step, as its evolved pool *is* the corpus. Both corpora are then minimised to remove inputs that are redundant for coverage.
3. **Minimization.** The raw corpus is minimised with **afl-cmin** to a coverage-preserving subset, a necessary step before seeding the AFL++ campaign.
4. **Seed-coverage benchmark (RQ1).** The minimised corpus is executed against the SUT and the total coverage it reaches is recorded.
5. **AFL++ campaign (6 h, RQ2).** The minimised corpus seeds a 6-hour AFL++ campaign, and coverage is logged over time.

4.3 Results

4.3.1 Corpus size

First we observe how large each tool’s seed corpus is, before and after minimisation. This is a basic sanity check and also gives context for the seed-coverage results of RQ1, since a larger corpus has more chances to cover more edges.

ELFuzz’s **produce** step runs each evolved generator for a certain amount of time, generating many seed inputs. As mentioned in Chapter 3.1, a single generator can emit an unbounded number of inputs, so ELFuzz naturally will have a much larger raw corpus than DLFuzz, which has no **produce** step, and whose evolved pool *is* the corpus. Also, the pool of inputs in DLFuzz will only grow if new inputs improve the coverage. This explains the massive disparity in corpus size before minimization.

After minimization with **afl-cmin**, seed corpora for all running configurations reduced in size drastically. DLFuzz’s jsoncpp corpus, shrinks from 116 to just 3 inputs (2.6%), showing that the directly generated inputs are highly redundant for coverage. Upon further inspection of the inputs, *none* of them were valid JSON, meaning none of the inputs were able to bypass the parser of the jsoncpp library. This indicates that the model used is very poorly conditioned on producing valid JSON, and does not “understand” the syntax. DLFuzz’s cpython3 corpus shrinks from 1,264 to 118 inputs (9.3%), better than jsoncpp, but still indicates a high level of redundancy. Only about a third of the inputs in the cpython3 corpus were valid Python, so the model is better at producing Python. See Section 4.3.5 for further discussion of input validity.

Comparing ELFuzz to DLFuzz, the former produces a much larger corpus, even after minimization. This would indicate that the model is much better at producing generators that can emit valid inputs, than it is at producing the inputs directly, which challenges our hypothesis. The generator abstraction appears to shield ELFuzz from the model’s weakness in directly generating valid inputs, since a generator can encode the structure of the input format once and emit many valid instances, whereas direct generation must get each input right on its own. This is a significant factor in the seed-coverage gap observed in RQ1.

SUT	ELFuzz		DLFuzz	
	Before	After	Before	After
jsoncpp	2,800,000	205	116	3
cpython3	880,000	822	1,264	118

Table 4.2: Seed-corpus size before and after minimisation with `afl-cmin`. “Before” is the raw corpus handed to `afl-cmin`: for ELFuzz the output of its 200-second `produce` step, for DLFuzz the evolved pool that doubles as its corpus. “After” is the minimised, coverage-preserving corpus that seeds the AFL++ campaign.

4.3.2 Coverage during pool evolution (DLFuzz)

Figure 4.1 traces the coverage held by DLFuzz’s pool over the course of evolution, for both SUTs. We show DLFuzz alone here, as its evolved pool is exactly the corpus it hands to AFL++, so the curve is a faithful picture of what the search produces. The equivalent curve for ELFuzz would not be comparable, because ELFuzz’s per-generation coverage reflects the coverage achieved by a small number of generated inputs during evolution. This is separate from the full dedicated production step after synthesis, which produces many more inputs per generator, thereby resulting in much different coverage.

These use the synthesis-phase (`getcov`) instrumentation, so they are not on the same scale as the `afl-showmap` seed-coverage figures reported in Section 4.3.3.

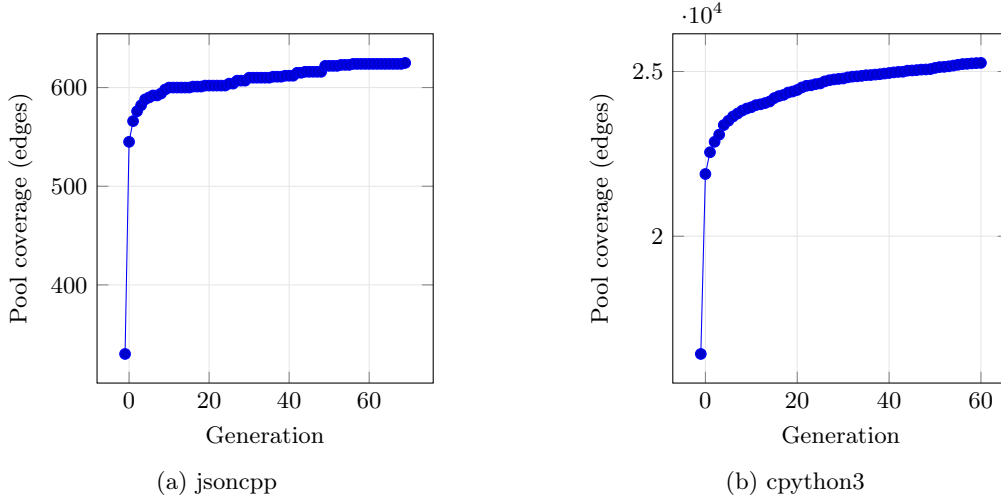


Figure 4.1: Coverage held by DLFuzz’s pool across generations (the running union of edges over all retained inputs, on the `getcov` instrumentation). Generation `-1` is the pre-generated initial seed set. Each later generation evaluates a batch of 1,000 candidates.

Both curves share the same overall shape: a steep initial climb followed by a long, decelerating tail. The first mutated generation alone accounts for most of the gain - on `jsoncpp` the pool jumps from 330 edges in the pre-generated seed set to 545 at generation 0, and on `cpython3` from 16,427 to 21,889 - after which each generation contributes progressively less. This is the expected signature of coverage-guided search: early mutations cheaply reach the easily-accessible parts of the SUT, while later ones must work harder for what remains.

jsoncpp plateaus much sooner and more completely than cpython3. On jsoncpp the curve flattens almost completely after generation 7 at 592 edges. It then rises to only 625 by generation 69, a gain of about 30 edges spread across the final sixty generations. However on cpython3, it continues to rise, but at a decelerating rate, until the end of the budget. At generation 56 it has reached 25,225 edges, and it is still rising at generation 60 at 25,261 edges. At roughly 25k edges it has reached only a fraction of the interpreter’s far larger reachable space, so a longer evolution budget would plausibly keep yielding new coverage, albeit at a diminishing rate. This is consistent with the difference in complexity between the two SUTs: jsoncpp is a much smaller simpler SUT as compared to cpython3, so its reachable frontier is smaller and easier to find, and the search saturates it sooner. A more complex SUT seems to be better suited to the evolutionary search of DLFuzz, as there is less chance of saturating the reachable frontier within a realistic budget, and more room for the search to find new coverage over time.

4.3.3 Seed coverage (RQ1)

Table 4.3 reports the coverage achieved by each tool’s minimised seed corpus on its own, measured with `afl-showmap`. This is the most direct measurement of seed quality.

SUT	ELFuzz	DLFuzz
jsoncpp	568	163
cpython3	19,634	9,790

Table 4.3: Seed-corpus coverage (edges covered by the minimised corpus, before any mutation fuzzing) for each tool, measured with `afl-showmap`. Higher is better.

On both SUTs the generator-synthesising pipeline produces a higher-coverage seed corpus than direct input generation: ELFuzz leads by roughly $3.5\times$ on jsoncpp (568 vs 163 edges) and $2\times$ on cpython3 (19,634 vs 9,790). This difference is partly due to the way in which the two seed corpora are generated. ELFuzz’s `produce` step generates a much larger raw corpus than DLFuzz, and DLFuzz’s seed corpus can only grow when new coverage is achieved. While improvements to coverage are part of the generator evolution process in ELFuzz, a single generator can achieve more coverage than was measured during evolution. With 880,000 / 2.8M inputs, ELFuzz’s seed corpus is allowed to risk some redundancy in exchange for sheer quantity. Also, as mentioned in Chapter 4.3.1, the model used is much better at producing generators that can emit valid inputs, than it is at producing the inputs directly, allowing ELFuzz’s seeds to reach past the initial parsing stage of the SUTs and cover more edges.

The difference in coverage is not just in quantity, as ELFuzz’s seed corpora are almost entirely *supersets* of our seed corpora. Table 4.4 shows the individual edges covered by each tool, and their relationship to each other.

SUT	ELFuzz	DLFuzz	Shared	ELFuzz-only	DLFuzz-only
jsoncpp	568	163	163	405	0
cpython3	19,634	9,790	9,755	9,879	35

Table 4.4: Edges covered by each tool’s minimised seed corpus and their overlap, measured with `afl-showmap` (so both tools sit in the same edge space). “Shared” is the intersection of the two edge sets; “ELFuzz-only” and “DLFuzz-only” are the edges unique to each tool. On both SUTs DLFuzz’s edges are almost entirely a subset of ELFuzz’s.

On jsoncpp every one of DLFuzz’s 163 edges is also covered by ELFuzz, which reaches 405 more; DLFuzz contributes *no* edge that ELFuzz misses. On cpython3, DLFuzz only achieves 35 unique edges not covered by ELFuzz, but ELFuzz covers 9,879 edges absent from DLFuzz’s corpus. Directly generated inputs are therefore exploring a subset of the generator-generated inputs, rather than a complementary part of the target. This is the strongest evidence against our hypothesis that direct generation would perform superior to indirect generation.

4.3.4 Downstream AFL++ coverage (RQ2)

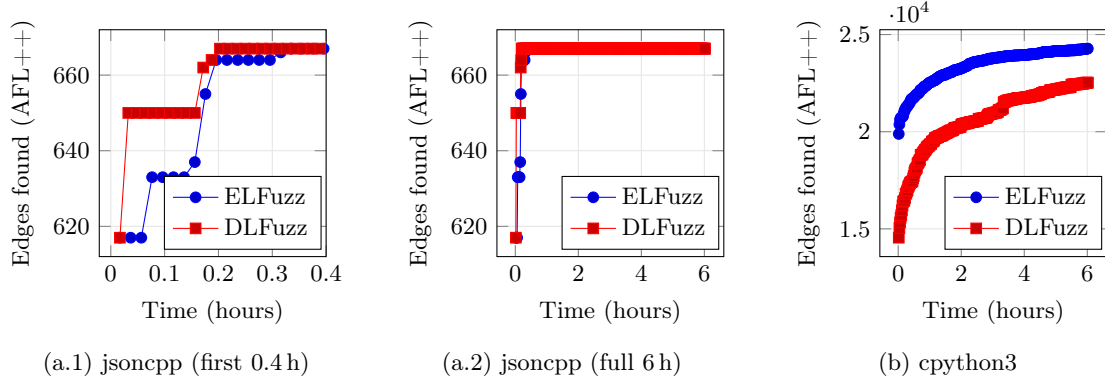


Figure 4.2: AFL++ coverage over time, seeded with each tool’s minimised corpus; campaigns run for 6 h. Panel (a.1) shows jsoncpp truncated to the first 0.4 h, where both tools reach the 667-edge reachable frontier within the first ~ 20 minutes and stay flat thereafter; (a.2) shows the same jsoncpp campaign over the full 6 h; (b) shows cpython3 over the full 6 h, where neither tool saturates.

SUT	Tool	Edges	Bitmap	Corpus	Execs	Crashes
jsoncpp	ELFuzz	667	12.74%	1311	166.9 M	0
	DLFuzz	667	12.74%	1468	141.7 M	0
cpython3	ELFuzz	24,277	18.20%	6043	3.31 M	0
	DLFuzz	22,523	16.88%	6636	3.33 M	0

Table 4.5: End-of-campaign AFL++ statistics (one repetition): edges found, bitmap coverage, final corpus size, total executions, and saved crashes. cpython3’s far lower execution rate (millions of execs, vs hundreds of millions for jsoncpp) reflects how much heavier each interpreter run is.

The two SUTs behave very differently. On jsoncpp the campaigns converge to the *identical* final coverage - 667 edges, 12.74% of the map - even though the ELFuzz seed corpus started out covering $3.5\times$ as many edges as the DLFuzz one. AFL++ saturates jsoncpp’s reachable frontier within the first few minutes (both curves are flat for almost the entire campaign), so the large seed-quality difference is erased entirely; it affects only how quickly the plateau is reached, not where it lands. On cpython3, by contrast, neither campaign comes close to saturation. They finish at 18.20% and 16.88% of a much larger map, and are both still climbing. Here the seed advantage persists: ELFuzz stays ahead throughout and ends with 24,277 edges compared to 22,523 edges for DLFuzz. The campaign narrows the relative gap (from $2\times$ at the seed stage to about 8% at the end) but does not close it.

4.3.5 Validity of the generated inputs

Of the 116 inputs in DLFuzz’s jsoncpp corpus, 0 were valid JSON. Upon inspection, apart from simple syntax errors, many of the inputs contained HTML tags, or JavaScript code, either directly in the input, in string literals, or after a `<file_sep>` token - this token is used by the model to indicate a separation between files. Clearly the model was not trained on raw JSON files, as they were often interpreted as part of a HTML or JavaScript context. This is a significant weakness for direct generation, as it means the model is not able to leverage its knowledge of the input format to produce valid inputs that can reach deeper into the SUT.

In contrast, 422 out of the 1,264 inputs in DLFuzz’s cpython3 corpus were valid Python. The model is much better at producing valid Python than valid JSON, which also can explain the good performance of ELFuzz, evolving python programs too. This allowed DLFuzz to produce a significant number of valid inputs that could reach deeper into the SUT, which is reflected in the higher seed coverage compared to jsoncpp. Still, only 33.4% of the seed corpus was valid Python, again severely limiting where in the SUT the inputs could reach, and therefore the coverage they

could achieve. This is a major factor in the seed-coverage gap between ELFuzz and DLFuzz on cpython3, as ELFuzz’s generator abstraction allows it to produce many valid inputs even if the model is not good at directly generating them.

4.4 Discussion

Here are some key takeaways from the results.

Using LLMs to generate seed inputs for fuzzing is less valuable for simpler SUTs, as demonstrated by jsoncpp. Even from 3 inputs after minimization, the SUT was saturated within 20 minutes. For cpython3, a much more complex SUT, neither ELFuzz nor DLFuzz saturated within the 6-hour budget. It is unlikely that cpython3 would saturate with an extended time budget (e.g. 24 or 48 hours), so there is more benefit to be had in using LLMs for fuzzing richer SUTs, such as other programming languages.

For cpython3, the quality of the seed corpus matters for the final coverage, as demonstrated by the persistent gap between ELFuzz and DLFuzz. AFL++ cannot exhaust the target within the budget, so it amplifies the seed corpus and preserves its quality difference throughout. This suggests that for realistic, large targets where saturation is unlikely, investing in a high-quality seed corpus is important for achieving better coverage.

The model’s ability to produce valid inputs is a critical factor for direct generation. DLFuzz’s jsoncpp corpus contained no valid JSON, which severely limited its coverage and AFL++’s ability to explore the SUT. In contrast, DLFuzz’s cpython3 corpus had a higher number of valid Python inputs, but still a low percentage overall. The generator abstraction of ELFuzz allows it to bypass the model’s weakness in generating many valid python programs, as a single generator can encode the structure of the input format and emit many valid instances.

With the evidence of these two SUTs, the directness hypothesis is not yet supported. These are early results, however - a single repetition, a small local model, and only two SUTs. A fuller comparison across more SUTs and repetitions is needed before drawing firm conclusions.

Chapter 5

Conclusion

The premise of this project was that using a large language model to generate fuzzing inputs directly, rather than to synthesise a generator that emits them, would yield a more diverse seed corpus and better fuzzing performance. The idea was to remove a layer of indirection, and utilise the LLM’s learned knowledge of the input format to write those inputs directly, in a syntactic and semantically aware manner. We implemented DLFuzz, which embodies this direct generation approach, and compared it against the original generator-synthesising ELFuzz on two systems under test: jsoncpp and cpython3 (Chapter 4). Prior work supports the broader idea: LLMs have been used to generate test inputs directly under a simple crash oracle with effective results [9, 12], and ELFuzz has shown that coupling an LLM with coverage feedback to evolve input *generators* is highly effective [11]. We combined these threads in DLFuzz, which uses an LLM to generate seed inputs directly and then evolves them under coverage guidance, and we hypothesised that this more direct route would yield a more diverse corpus and, in turn, better fuzzing performance.

On the evidence gathered, that hypothesis is not supported. Across the two systems under test we evaluated, the original generator-synthesising ELFuzz matched or beat DLFuzz on every end-to-end measure (Chapter 4). Its seed corpora reached more coverage on their own (568 versus 163 edges on jsoncpp, 19,634 versus 9,790 on cpython3), and the edge sets they covered almost entirely subsumed those of DLFuzz, rather than a complementary part of the target. Downstream, the gap was erased on the small jsoncpp - where AFL++ saturated the reachable frontier within minutes regardless of the seed - but it persisted on the larger cpython3, where ELFuzz stayed ahead for the whole campaign. We did not observe the diversity advantage that motivated the approach.

Several factors plausibly contribute, and they bound how strongly this result should be read. The most immediate is the generator itself. Hardware constraints limited us to a small local model, Qwen2.5-Coder-1.5B [46], where the original ELFuzz experiments used the far larger CodeLlama-13B [39, 11]. The cost of this was visible directly in the inputs: not one of the jsoncpp pool’s inputs was well-formed JSON, and only about a third of the cpython3 pool compiled as valid Python. A model that cannot reliably emit a complete, valid instance of a simple format will struggle to produce the structurally varied inputs that the hypothesis claims will be more effective. The indirection of synthesising a generator may in fact be what shields ELFuzz from this weakness - a generator can encode the format’s structure once and emit many valid instances, whereas direct generation must get each input right on its own.

A second factor is the evolution rule. DLFuzz curates its pool purely on coverage, retaining an input only when it contributes a new edge. This is a strict criterion, and it interacts poorly with the way ELFuzz appears to win: its **produce** step samples 1 million candidate inputs from the evolved generators, and that sheer quantity - not just per-input quality - is a large part of where its coverage comes from. A evolution strategy that entirely optimises for coverage growth (DLFuzz’s jsoncpp corpus minimised to just three inputs) will discard the redundant-but-varied volume that benefits the downstream mutation fuzzer. Optimising for coverage alone may therefore be the wrong objective for a corpus that is about to be handed to AFL++.

5.1 Future Work

These are early results, based on a single run, a small model, and only two systems under test (Chapter 4), so there is plenty of room to build on them. The most pressing limitation is the model itself, and it showed in the outputs: very few of the generated inputs were even valid, so most of them only exercised the target’s parser and never reached the logic behind it. The natural next step is to repeat the experiment with a larger, more capable model on hardware that can host it, and to run the campaigns for longer, such as the 24 hours used in the original ELFuzz evaluation [11] rather than the six we used. That would show whether the validity problem is fundamental to direct generation or simply an artefact of model size, and give any real difference in seed quality more time to show. Input validity is worth tackling head-on as well, since getting the model to produce valid inputs is the key to exercising more than just the parser. Constraining the model so that it can only emit tokens that are valid in the target format, or using a model trained on that format, are both promising ways to do this.

It would also help to test on more targets, and on richer ones. jsoncpp turned out to be too simple to be informative, as AFL++ exhausted its reachable code almost immediately for both tools, so the quality of the seeds made no difference downstream. Larger, more complex targets such as programming-language interpreters and compilers are a much better test of the idea, because their input space is nowhere near exhausted within a realistic time budget. The way DLFuzz curates its pool is also worth revisiting. It keeps an input only when it adds new coverage, but part of ELFuzz’s strength seems to come from sheer quantity, producing around a million inputs before minimising. A more permissive rule that keeps a larger, more varied pool, or that rewards structural and semantic diversity directly rather than coverage alone, may suit a downstream mutation fuzzer better than the strict consolidation we used here.

Chapter 6

Declarations

6.1 Generative AI Usage

Several large language models were used during this project: GPT-5, Gemini 3.1, and Claude Opus 4.8. Their use is described below.

In the background section, Claude Opus 4.8 was used to help find potentially relevant references and papers to cite. Every suggested source was checked by hand before being included.

During development, Claude Opus 4.8 was used in a planning capacity to help blueprint the design and make implementation decisions. It also helped to debug difficult concurrency and memory-related bugs.

The project itself uses LLMs as a core component: Claude Opus 4.8 generates the initial seed inputs, and Qwen2.5-Coder-1.5B drives the evolution loop.

In the evaluation, Claude Opus 4.8 was used to write scripts that take the raw outputs from runs on the cluster and turn them into the tables and figures presented in this report.

For the write-up, GPT-5 was used to proofread the report and identify unclear passages.

6.2 Ethical Considerations

This project raises no ethical concerns. It evaluates fuzzing techniques on open-source software and does not involve human participants, personal data, or sensitive systems.

6.3 Sustainability

This project raises no sustainability concerns beyond the energy used to run the experiments, which were carried out on a single workstation over a bounded time budget.

6.4 Availability of Data and Materials

All source code is available at <https://github.com/FaresShehata/final-year-project>, and the experimental results are available at <https://github.com/FaresShehata/final-year-project/tree/main/results>.

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